

May 2025 | eBook

Set a Course for Kubernetes Success

Simplify the delivery of modern apps at scale

NUTANIX

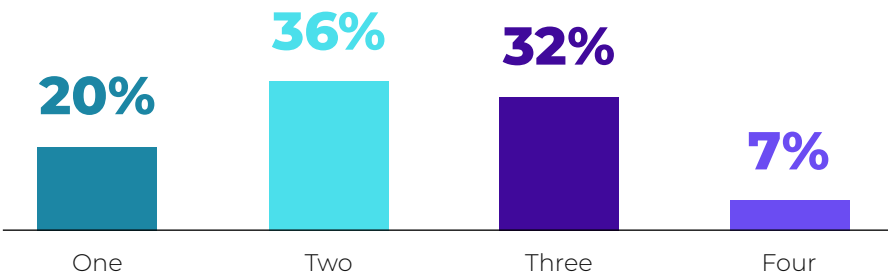


Enterprises Need to Leverage Kubernetes Everywhere

Many enterprises have focused their initial Kubernetes efforts on the cloud. Cloud Kubernetes services offer numerous benefits. They can be easy to set-up and use, include basic management and support capabilities, and eliminate some operational tasks.

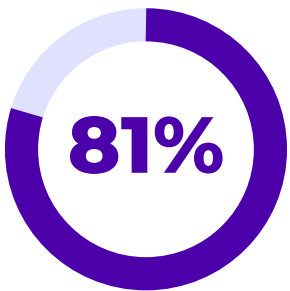
However, enterprises are hitting roadblocks when it comes to running Kubernetes in production at scale. Many find that cloud Kubernetes services don't eliminate the need for Kubernetes in the datacenter or at the edge. In a 2025 survey, the majority of respondents reported running two or more Kubernetes environments.

Number of Kubernetes Environments Deployed

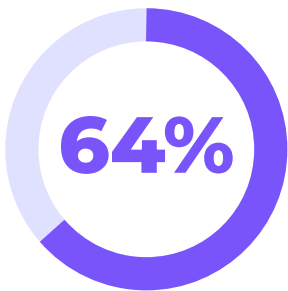


Source: 2025 Enterprise Cloud Index report by Vanson Bourne

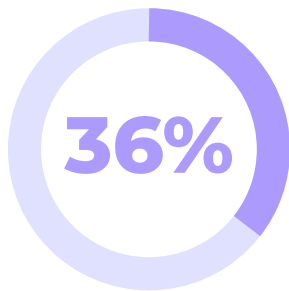
How can you run Kubernetes across diverse environments that may include datacenters, public clouds and edge while simplifying operations and gaining a competitive advantage? And how can you unify your tools to standardize Kubernetes deployment and simplify management?



feel their current IT infrastructure requires improvement to fully support cloud native applications and containers



find cloud native and container application development challenging



feel they do not have all of the necessary skills needed to support cloud native applications and containers

Source: 2025 Enterprise Cloud Index report by Vanson Bourne

This eBook explains the pluses and minuses of cloud Kubernetes services and what capabilities you need to get the most out of your investment. You'll also discover how Nutanix Kubernetes Platform (NKP) provides enhanced control, flexibility and visibility to deliver production applications in the cloud, on-premises and at the edge with speed and scale.

Cloud Kubernetes Services: Pros and Cons

Kubernetes has rapidly become the preferred way to build new software experiences and modernize existing applications, making it far simpler to run applications at scale and across different environments. With Kubernetes, you can host workloads in a single cloud, across multiple clouds, in datacenters, and at the edge. Its capabilities lend themselves equally well to public cloud, hybrid cloud and multicloud strategies. Many enterprises take advantage of cloud Kubernetes services, such as Amazon Elastic Kubernetes Service (EKS) and Azure Kubernetes Services (AKS), to help build, deploy and manage containerized applications

Benefits of cloud Kubernetes services

Low barrier to entry. Getting up and running on a cloud Kubernetes service is relatively easy so you don't need much expertise upfront. Cluster deployment is automated by the provider and the cluster is managed for you.

Minimize operational tasks. Some operational tasks, such as infrastructure management, are outsourced to the cloud provider. That means more time to dedicate to more important projects.

A broad catalog of cloud services. Enterprises can easily take advantage of cloud databases, developer tools and other services to simplify application development. Self-service access to a broad catalog lets you get started prototyping, building and deploying new digital services quickly.

Management and support. Patching, updates, fixes, node management, and other types of lifecycle management are included. This reduces the operational burden on your team.

However, many new challenges can come to light when running rapidly expanding Kubernetes operations in production using these cloud services.

Pain points associated with cloud Kubernetes services

No consistency across environments. Each cloud platform has its own set of tools and capabilities to deploy and operate Kubernetes. If you're using EKS, the tools, services and interfaces you're using are specific to EKS.

If you're taking a hybrid cloud or multicloud approach, you may end up with different tools, services and interfaces in every environment.

Lack of centralized observability and management. As your cloud-based Kubernetes operations expand, it can become difficult to keep track of everything. One team might be building its stack on EKS while another is building on AKS.

Even if you stick to a single cloud, clusters may exist that are outside of your control. You need centralized visibility into your entire fleet of clusters and a consistent way to manage and obtain insights.

No multitenancy. When multiple users and groups access the same Kubernetes clusters without multitenancy controls, resource sharing, identity controls and access management can become a challenge. This is a major contributor to cluster sprawl.

Insufficient application management. Managing a production-ready environment requires more than just Kubernetes. It also requires tools for logging, monitoring and tracing. Most cloud Kubernetes services do little to help you get these capabilities up and running.

If you have multiple clusters across several clouds, continuous deployment becomes a headache and causes inconsistencies that introduce performance and reliability issues as well as security risks.

Inadequate cost visibility and management. Cluster sprawl is real. While cloud providers offer cost management tools, it's important to have real-time cost granularity, including deep awareness of which teams are using which services, what each application costs to run, and where there are opportunities for optimization.

Lack of centralized governance and policies. If you're not able to create policies to govern Kubernetes usage by team, product or environment, you'll never gain the level of control over your operations that's required to succeed.

Not enough expertise. Some organizations that use cloud Kubernetes services don't have the experience to deploy and run applications efficiently. It's difficult to recruit and hire people with the proper skills or train the people you have.

Key Capabilities for Kubernetes

Understanding the pain points and gaps associated with cloud Kubernetes services can help your team identify opportunities to streamline processes, reduce overhead and get the most out of your Kubernetes investment.

Following are critical insights you should consider when choosing and deploying Kubernetes solutions in the cloud, on-premises and at the edge.

Consistency across environments

Can you deploy and manage all clusters in all environments using a single set of tools?

Observability

Do you have visibility of all your Kubernetes clusters with the metrics you need to monitor and manage them effectively?

Financial visibility

Do you have visibility of Kubernetes costs at the right granularity for your business needs?

Application management

Can you deploy and manage applications efficiently across different environments?

Authorization and authentication

Do you have a simple and secure way to authenticate users and control access to services everywhere?

Lifecycle operations

Can you consistently deploy, scale, upgrade, and manage all your Kubernetes clusters?

Open platform

Does your solution integrate with upstream and open-source services for security, storage and networking?

Portability

Can you move applications across cloud, datacenter and edge locations with minimum friction?

Workload coverage

Does your solution support the workloads that matter to your business – including AI – with the necessary scale, security and resilience?

Cloud-native expertise

Does your chosen vendor have the expertise to accelerate your Kubernetes journey?

Full support

Does the solution include support that you can depend on when deploying business-critical apps and services?

Governance

Can you enforce consistent policies for cluster deployment, management, access, and security?

Reap the Full Benefits of Kubernetes with Nutanix

While cloud Kubernetes services seem like a no-brainer, they leave a lot to be desired when it comes to enterprise requirements. How can you run Kubernetes efficiently in the cloud and on-premises? How do you unify the experience so you can build, deploy and manage Kubernetes and cloud-native applications seamlessly across different environments?

Nutanix Kubernetes Platform (NKP) reduces operational complexity and establishes consistency across any environment. NKP saves time and resources, enabling your team to operationalize production-ready environments in minutes rather than weeks or months.

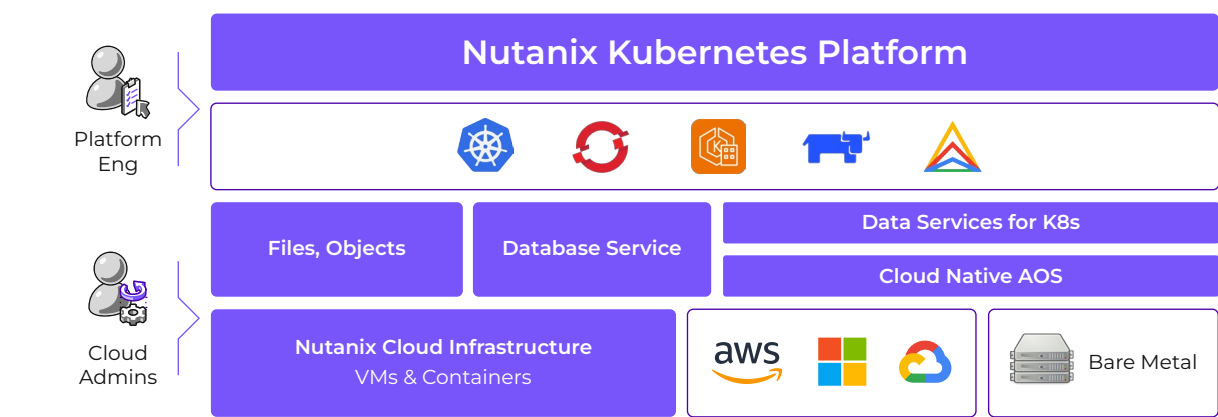
This enables you to overcome critical challenges that are common in Kubernetes environments, including security, observability, reliability, upgradeability, data protection, policy management, and governance.

Key benefits

- **Turnkey.** All the components needed for production-ready Kubernetes in a fully integrated solution.
- **Flexible deployment options.** Deploy in the public cloud, on-premises and at the edge with or without Nutanix Cloud Infrastructure.
- **Full-featured management.** Control operations using our intuitive graphical interface or integrate easily with your existing operations using our command-line interface or GitOps automation.
- **Open.** Built from upstream Cloud Native Computing Foundation (CNCF) projects that are fully integrated and validated. Any component is easily replaced so you're not locked-in.
- **Military-grade security.** Exceeds CISA/NSA Kubernetes security hardening guidelines and can be configured to meet defined security standards. Air-gapped deployments are supported.

Featured capabilities

- **Fleet management.** Nutanix simplifies Kubernetes management with consistent security, visibility, automation, and governance features across all clusters.
- **Cluster Life Cycle Manager (LCM).** Nutanix reduces the operational burden and enhances cluster reliability and efficiency with:
 - Easy cluster upgrades
 - Built-in cluster autoscaler
 - Self-healing cluster resilience
- **Nutanix Data Services for Kubernetes (NDK).** Simplify and unify cloud-native operations with enterprise data services for containerized apps.
- **NKP Insights.** This unique capability from Nutanix collects events and metrics and applies AI for anomaly detection with root-cause analysis.
- **AI Navigator.** An interactive chatbot with deep domain expertise that simplifies management and helps your team solve complex problems.



With NKP, a management cluster encapsulates all the capabilities necessary for a complete cloud-native environment and provides the NKP control plane. Managed clusters are deployed and managed centrally by the NKP control plane.

The following table explains all the key capabilities you'll need to succeed with Kubernetes.

Consistency across environments A single management plane and toolset spans cloud, datacenter and edge deployments.	Observability Centralized visibility of Kubernetes workloads with metrics to manage service availability.	Financial visibility Granular real-time visibility into Kubernetes costs (including chargeback) and infrastructure insights to optimize cloud infrastructure spend and improve workload utilization.
Application management Built-in cloud-native application deployment tools facilitate zero downtime deployments.	Authorization and authentication Single sign-on and RBAC to identify users, configure access and perform compliance checks.	Lifecycle operations Automation to consistently deploy, scale, upgrade, and manage multiple clusters.
Complete and open platform Integration with other upstream and open-source services for security, storage and networking, with end-to-end automation out-of-the-box.	Portability Build, deploy and manage Kubernetes and cloud-native applications on cloud, datacenter and edge or any combination.	Workload coverage Run complex, mission-critical business services, data-rich apps, streaming analytics, and machine learning use cases with enterprise-grade requirements of scale, security and resilience.
Cloud-native expertise Nutanix has the experience and expertise to help your team come up to speed and succeed with Kubernetes.	Industry-leading support Nutanix has maintained a 90+ Net Promoter Score (NPS) average for nearly a decade, ranking high in the technology industry for customer loyalty and satisfaction.	Governance Centralized governance and policy administration for consistency across cluster deployment, management, access, and security.



Learn more about Nutanix Kubernetes Platform on our website or take it for a free test drive.

[Learn More](#)

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